Computer Engineering Department
Cairo University

Operating Systems
Lab 1
Bash Scripting

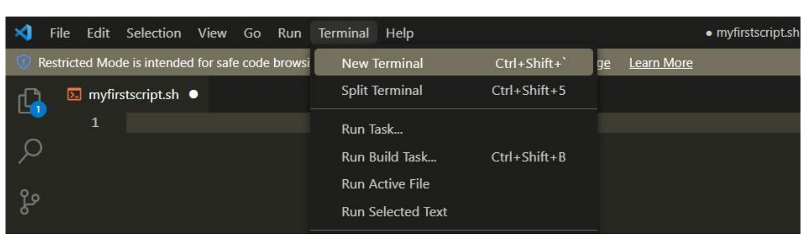
Hints:

- All commands you need are already given .
- You might need options that we didn't mention, so use the 'man' command to get help and to know what other options are there.
- In order to get the full mark, you need to do all the steps using commands only and each step in one line.
- If you get stuck in a command, pass it (do it using gui) , and continue to do the rest , then try it at the end.

Script Steps:

- If you are using WSL2 you will not find the Desktop directory by default. You can create it by Opening a terminal and typing
  - `mkdir ~/Desktop`
  - `cd ~/Desktop`
- Also, if you are using WSL2. Open vscode in windows and download WSL extension.

- type this command
  - `code ~/Destop/` → this will open a GUI editor in the background.
  - Open a new terminal in VSCode if you don't see an already open one.



- In the terminal type
  - `touch myfirstscript.sh`
- First line of the script should direct the terminal into which interpreter to use to interpret your file, here we use the **bash interpreter** and it's located under /bin. The **#!** symbol is the directive.
  Type: "**#! /bin/bash**" at the beginning of your file.
- In the line after it, use **echo command** to write your name to the terminal
- From the terminal, change the permissions of the file to be executable, as mentioned
  - `chmod u+x myfirstscript.sh`
- Now run your script from the terminal typing either of the following
  - `./myfirstscript.sh`
  - `source ./myfirstscript.sh`
- Now your script is ready for the real requirement, understand the requirement and add the command that executes what is required to the file: (Hint, run your file after each command)
- Download the two files uploaded with the requirements "**words.txt"** & "**numbers.txt"** and add them to your ~/Desktop

Requirement Steps:

1. Navigate to your ~/Desktop directory (don't include the navigation command in the script just do it in your terminal)
2. Delete the directory Lab1 with its subdirectories and files (if it exists but no need to do any validations).
3. Create a new directory called Lab1.
4. Copy the files "**words.txt"** & "**numbers.txt"** into Lab1 directory (they should by already in your ~/Desktop if you followed the Script Steps section above).

5. Create a new file named “**MergedContent.txt**” in Lab1 directory that contains the content of the words & numbers merged **<u>side by side</u>**.
6. Display in the terminal the first **<u>three</u>** lines in the file **“MergedContent.txt”.**
7. Sort the “**MergedContent.txt”** and save your result in the file “**SortedMergedContent.txt”** in the same folder.
8. Display on the screen the following message “**The sorted file is :**”
9. Display in the terminal the content of “**SortedMergedContent.txt”.**
10. Prevent **<u>anyone</u>** from reading the “**SortedMergedContent.txt”.**
11. Display the contents of the “**MergedContent.txt”** after removing the duplicate lines as we did in the section. (You do not need to remove the duplicate lines from the actual file. Just display them without duplicates also the output doesn’t have to preserve original lines order).
12. Try to replace all small letters in “**SortedMergedContent.txt”** with capital ones in new file **“CapitalSortedMergedContent.txt”.**
13. Display a message on the screen explaining what happened in the last step and why?
14. Solve the problem in (13) and re-run (12) again.
15. Find lines starting with “w” and ending with a number in “**MergedContent.txt”** (display the line numbers as well)**.**
16. Replace every occurrence of “**i**” in “**MergedContent.txt”** with “o” and save it to “**NewMergedContent.txt”** in the same folder.
17. Display both files **“MergeContent.txt”** & **“NewMergeContent.txt”** side by side on the terminal.

SUBMISSION:

- The shell file should be named: "req_1_firstName_secondName_ID.sh"
- Functions should be named exactly like stated in the requirements above
- Do not add additional input parameters or swap the order of the parameters